Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/13/2016
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Colin, Catherine Ph.D. (Institut de Mécanique des Fluides de Toulouse)		
Grant/Contract No.:	NNX15AK16G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	EDITOR'S NOTE (5/17/2016): Continuation of "Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry" with same Principal Investigator, Dr. Jungho Kim, grant # NNX12AF31A. Added to Task Book when received information. This effort will be conducted with the European Space Agency (ESA) in one of the facilities/test articles that they are developing. The work performed under previous grants has demonstrated that detailed time and space resolved measurements of the heat transfer distribution can be obtained in flow boiling along with flow visualization using IR (infrared) thermometry. This capability has greatly increased our understanding of the heat transfer within the various regimes of flow boiling, and can be used to verify models that have been proposed for these sub-regimes and to develop new models as needed. The existing test apparatus using this technique allowed us to obtain local heat transfer distributions before, during, and		

	after passage of a vapor bubble slug as it rose in a vertical tube under the high-g, 1-g, and low-g conditions produced by parabolic aircraft. The bubble shape and behavior were also studied under those conditions. Tests were conducted using a flow boiling rig which coupled flow visualization with measurements of the heat transfer and liquid film thickness. The bubble drift velocity was shown to dictate the bubble shape as well as the relative influences of thin film evaporation and turbulent wake heat transfer. When present, vortices shed behind the bubble tail created peaks in the heat transfer coefficient whose magnitude decreased with increasing liquid Reynolds number and acceleration. The vortex frequency was found to be linearly dependent on plunging velocity. A criteria for vortex incipience was proposed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	Quantum Dots (QD) Preliminary work was completed to evaluate whether or not quantum dots could be used to measure local heat transfer distributions. This is a potentially low-cost technique that may enable use of much smaller CCD/CMOS cameras in place of bulky and expensive IR cameras, and may be more appropriate for a space-based experiment where space and power are severely limited. Quantum dots are nanoscale semiconductor crystals that fluoresce when excited by blue/UV light. The emission color depends on the dot size with larger dots emitting at longer wavelengths. The intensity of emission decreases and the peak in the emission shifts to longer wavelengths with increasing temperature. Preliminary experiments at UMD (University of Maryland) since October 2014 have indicated that quantum dots show promise as a method to measure wall temperature distributions from which heat transfer distributions can be derived. Changes in overall emission intensity with a temperature resolution of <0.2°C could be measured using 12-bit CCD cameras. However, key challenges to this technique were found to include uniform deposition of quantum dots, stability of illumination, and subpixel alignment of the camera and target viewing area throughout the calibration and data acquisition process. Given the high cost of quantum dots and the relatively low sensitivity to temperature, it was decided to pursue an alternate technology based on Temperature Sensitive Paints (TSPs). Temperature Sensitive Paints (TSPs) The principle of using TSPs to measure temperature is similar to that of QDs. When excited by blue or UV light, the emission intensity decreases with temperature and with oxygen concentration. The sensitivity to oxygen is minimized by encapsulating the TSP within an impermeable binder (polyacrilic). The advantage of TSPs over QDs are primarily their lower cost and higher sensitivity. For the ruthenium based TSPs, a temperature sensitivity of about 4%/K was observed. The large sensitivity to temperature, however, means the maximum temperature at which TSPs can be used is limited (Tmax~120°C for the ruthenium based TSPs we have been using). We have been able to deposit a thin layers of TSP onto a plastic PET substrate which was then bonded to an opaque layer of polyimide containing electrically conductive carbon. The polyimide was resistively heated and the temperature of the TSP monitored as the heat flux was increased during boiling of HFE7000. Lower temperatures are seen where bubbles nucleate and depart from the surface. Natural convection occurs between nucleation sites. We have developed algorithms to compute the local heat flux from this temperature data. The TSP technique is being developed further to test its sensitivity to pressure, time, illumination intensity (high intensities can cause photobleaching). Once the technique is qualified, we will use it to study pool boiling heat transfer at high heat fluxes (CHF) as well as modify the flow boiling test section so the TSP can be used in place of the IR technique.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Articles in Peer-reviewed Journals	Moaveni S, Kim J. "An inverse solution for reconstruction of the heat transfer coefficient from the knowledge of two temperature values in a solid substrate." Inverse Problems in Science and Engineering. Published online: 21 Mar 2016. http://dx.doi.org/10.1080/17415977.2016.1161035 , Mar-2016
Articles in Peer-reviewed Journals	Scammell A, Kim J, Magnini M, Thome JR. "A study of gravitational effects on single elongated vapor bubbles." International Journal of Heat and Mass Transfer. 2016 Aug;99:904-17. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2016.04.034 , Aug-2016
Papers from Meeting Proceedings	Scammell A, Kim J. "A study of Taylor bubble heat transfer mechanisms under variable gravity conditions." First Pacific Rim Thermal Engineering Conference, Hawaii, March 13-17, 2016. First Pacific Rim Thermal Engineering Conference, Hawaii, March 13-17, 2016. Paper PRTEC-14462. , Mar-2016